5.2.2 Distinctive behavior of lithium

The distinctive behaviour of Li^+ ion is due to its exceptionally small size, high polarising power, high hydration energy and non availability of d-orbitals.

Table 5.5 Comparison of properties of lithium with other elements of the group:

Lithium	Other elements of the family
Hard, high melting and boiling point	Soft and Lower melting and boiling point
Least reactive (For example it reacts with oxygen to form normal oxide, forms peroxides with great difficulty and its higher oxides are unstable)	More reactive
Reacts with nitrogen to give Li_3N $6Li + N_2 \rightarrow 2Li_3N$	No reaction
Reacts with bromine slowly	React violently
Reacts directly with carbon to form ionic carbides. For example $2Li + 2C \rightarrow Li_2C_2$	Do not react with carbon directly, but can react with carbon compounds. $2Na + C_2H_2 \rightarrow Na_2C_2 + H_2$
Lithium nitrate decomposes to give an oxide	Decompose to give nitrites

Table 5.6 Similarities between lithium and Magnesium

S.No.	Properties
1	Both lithium and magnesium are harder than other elements in the respective groups
2	Lithium and magnesium react slowly with water. Their oxides and hydroxides are much less soluble and their hydroxides decompose on heating.
3	Both form a nitride, Li_3N and Mg_3N_2, by direct combination with nitrogen
4	They do not give any superoxides and form only oxides, Li_2O and MgO
5	The carbonates of lithium and magnesium decompose upon heating to form their respective oxides and CO_2.
6	Lithium and magnesium do not form bicarbonates.
7	Both LiCl and $MgCl_2$ are soluble in ethanol and are deliquescent. They crystallise from aqueous solution as hydrates, $LiCl{\cdot}2H_2O$ and $MgCl_2{\cdot}8H_2O$

Diagonal Relationship:

Similarity between the first member of group 1 (Li) and the diagonally placed second element of group 2 (Mg) is called diagonal relationship. It is due to similar size (r_{Li^+} = 0.766 Å and $r_{Mg^{2+}}$ = 0.72 Å) and comparable electronegativity values (Li = 1.0; Mg = 1.2).

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